

GRIDINO, Russian Federation

WMO: 224220

Lat: 65.900N

Lon: 34.767E

Elev: 39

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 82-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-9.6	-17.4	2.1	-14.0	-12.6	2.8	-9.5	35.4	16.4	30.3	22.2	2.1	320	0.772

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	71.8	60.9	67.7	59.3	64.7	57.6	62.7	69.6	60.4	66.1	58.8	63.3	8.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	76.7	65.0	58.0	72.0	62.6	56.5	68.1	60.8	28.3	69.0	26.7	66.1	25.6	63.3	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 31.0	26.4	21.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-20.5	67.1	7.3	4.0	-25.7	69.9	-30.0	72.3	-34.1	74.5	-39.4	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	10.2	15.1	22.6	29.2	39.7	49.9	56.7	53.7	46.0	37.7	24.3	16.2
	DBStd	17.58	12.71	11.89	9.14	8.00	7.69	7.59	5.20	5.50	5.33	6.25	8.77	10.60
	HDD50	6482	1233	977	849	623	335	95	5	25	142	381	770	1047
	HDD65	11483	1698	1397	1314	1073	784	454	263	353	569	846	1220	1512
	CDD50	484	0	0	0	0	16	93	214	138	23	0	0	0
	CDD65	11	0	0	0	0	0	3	7	1	0	0	0	0
	CDH74	74	0	0	0	0	4	18	51	1	0	0	0	0
	CDH80	10	0	0	0	0	0	1	9	0	0	0	0	0
Wind	WSAvg	8.6	8.6	7.8	8.3	8.3	7.9	8.4	8.0	9.1	8.8	10.1	9.2	9.0
Precipitation	PrecAvg	18.1	1.2	1.0	0.9	0.9	1.5	1.7	2.2	2.3	1.8	1.8	1.6	1.4
	PrecMax	26.9	2.0	1.6	1.8	1.7	2.8	3.3	4.3	6.4	3.9	4.2	2.8	3.0
	PrecMin	12.2	0.3	0.2	0.2	0.3	0.4	0.4	0.2	0.6	0.3	0.6	0.1	0.3
	PrecStd	3.5	0.4	0.4	0.4	0.3	0.6	0.7	1.1	1.3	0.9	0.9	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	38.4	41.1	54.6	71.5	76.3	81.6	72.3	66.1	52.9	42.3	37.4
		MCWB	34.6	36.1	35.7	45.0	58.3	61.0	63.5	62.6	56.9	47.8	40.3	35.5
	2%	DB	33.9	35.6	38.6	48.1	61.5	69.5	72.9	67.4	60.1	50.3	39.5	35.0
		MCWB	32.3	33.6	34.8	41.3	50.9	58.3	62.0	60.3	54.9	46.8	38.2	33.5
	5%	DB	30.8	33.8	36.5	44.6	56.3	65.2	68.8	64.6	56.4	48.1	36.7	33.2
		MCWB	29.5	32.2	33.8	38.7	48.2	56.3	60.5	58.7	52.5	45.4	35.3	31.9
	10%	DB	27.7	31.3	34.4	41.2	50.6	61.3	65.5	62.1	53.7	45.9	35.1	31.1
		MCWB	26.6	30.1	32.3	36.7	44.6	54.7	58.8	57.9	51.1	44.2	34.1	29.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	36.5	37.3	46.2	58.3	62.8	66.4	64.7	58.2	50.0	41.2	36.0
		MCDB	36.1	37.9	39.6	51.4	69.7	73.0	72.7	70.2	63.6	51.5	41.9	37.1
	2%	WB	32.4	34.2	35.4	41.9	52.5	59.6	63.6	61.4	55.4	47.2	38.3	33.7
		MCDB	33.9	35.3	37.5	47.9	60.3	67.5	71.3	65.5	59.6	48.9	39.5	34.6
	5%	WB	29.6	32.3	33.9	39.3	48.1	57.1	61.3	59.6	53.0	45.7	35.7	32.0
		MCDB	30.9	33.4	36.1	43.6	54.6	63.4	67.7	63.2	55.3	47.4	36.3	32.9
	10%	WB	26.6	30.1	32.5	37.0	44.8	55.0	59.3	58.3	51.3	44.3	34.2	29.9
		MCDB	27.5	31.3	34.5	40.7	50.3	60.5	64.2	61.6	53.5	46.1	35.0	31.1
Mean Daily Temperature Range	5% DB	MDBR	11.4	10.2	11.1	11.4	10.3	10.4	9.7	8.3	8.0	7.1	8.5	11.2
		MCDBR	11.4	8.4	10.9	15.0	20.2	18.2	16.6	13.6	12.6	9.6	7.6	8.7
		MCWBR	11.0	7.9	8.5	10.2	12.1	10.2	8.6	8.1	8.4	7.0	7.3	8.6
	5% WB	MCDBR	11.8	8.0	9.8	14.0	18.8	15.2	14.8	11.2	11.3	8.2	6.8	8.3
		MCWBR	11.5	7.6	8.0	10.5	12.1	9.4	8.7	7.4	8.3	6.7	6.8	8.5
Clear-Sky Solar Irradiance	taub	0.217	0.275	0.305	0.337	0.351	0.347	0.374	0.350	0.326	0.297	0.349	0.358	
	taud	1.951	2.077	2.139	2.253	2.420	2.488	2.450	2.489	2.528	2.349	2.584	2.604	
	Ebn at Noon	70	180	237	261	271	274	262	257	238	180	73	22	
	Edh at Noon	6	18	28	33	31	30	30	26	20	14	6	3	
All-Sky Solar Radiation	RadAvg	26	187	663	1172	1496	1755	1603	1121	623	214	42	4	
	RadStd	3	34	68	91	111	159	178	109	64	30	6	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	-272	-360	N/A	N/A

Nomenclature: See separate page